

Digital Forensics

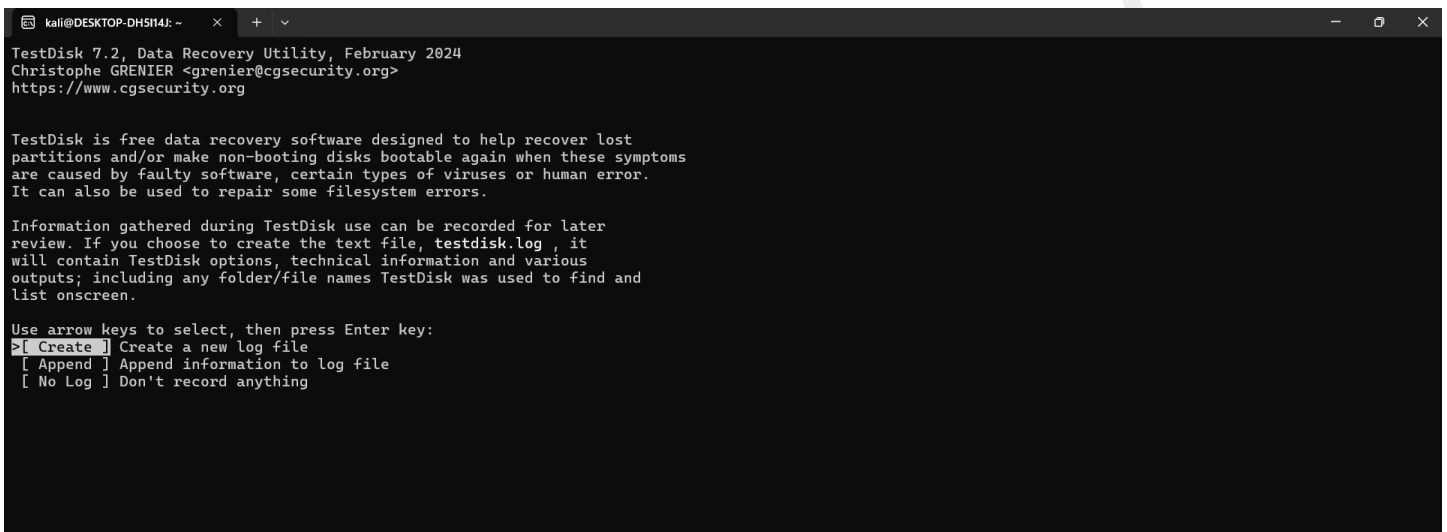
Ex.No:2 Recover deleted or damaged files from a storage device using Test Disk

Date:

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation



```
kali@DESKTOP-DH5H4J: ~
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log, it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
>[ Create ] Create a new log file
[ Append ] Append information to log file
[ No Log ] Don't record anything
```

- Choose Create to instruct Testdisk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or you execute TestDisk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk selection

All hard drives should be detected and listed with the correct size by TestDisk:

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TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk /dev/nvme0n1 - 512 GB / 476 GiB - PC SN740 NVMe WD 512GB

>[Proceed] [Quit]

Note: Serial number 23452R806811
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

[X](#) If available, use raw device `/dev/rdisk*` instead of `/dev/disk*` for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.

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Disk /dev/nvme0n1 - 512 GB / 476 GiB - PC SN740 NVMe WD 512GB

Please select the partition table type, press Enter when done.
[Intel ] Intel/PC partition
>[EFI GPT] EFI GPT partition map (Mac i386, some x86_64 ...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection

Hint: EFI GPT partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current partition table status

TestDisk displays the menus (also see [TestDisk Menu Items](#)).

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Disk /dev/nvme0n1 - 512 GB / 476 GiB - PC SN740 NVMe WD 512GB
CHS 488386 64 32 - sector size=512

> [Analyse] Analyse current partition structure and search for lost partitions
[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options] Modify options
[Quit] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

- Use the default menu "Analyse" to check your current partition structure and search for lost partitions.
- Confirm at Analyse with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.

```
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Disk /dev/nvme0n1 - 512 GB / 476 GiB - CHS 488386 64 32
Current partition structure:
Partition      Start      End      Size in sectors
1 P EFI System    2048      616447    614400 [EFI system partition]
No FAT, NTFS, ext2, JFS, Reiser, cramfs or XFS marker
2 P MS Reserved   616448    878591    262144 [Microsoft reserved partition]
2 P MS Reserved   616448    878591    262144 [Microsoft reserved partition]
Warning: number of heads/cylinder mismatches 255 (NTFS) ≠ 64 (HD)
Warning: number of sectors per track mismatches 63 (NTFS) ≠ 32 (HD)
3 P MS Data       878592    778887167 778008576 [Basic data partition] [OS]
Warning: number of heads/cylinder mismatches 255 (NTFS) ≠ 64 (HD)
Warning: number of sectors per track mismatches 63 (NTFS) ≠ 32 (HD)
4 P MS Data       778887168 783081471 4194304 [Basic data partition] [New Volume]
5 P EFI System    783083520 785084415 2000896
6 P Linux fileysys. data 785084416 942192639 157108224
7 P Linux Swap    942192640 950857727 8665088
8 P Windows Recovery Env 950857728 953268223 2410496
9 P Windows Recovery Env 953268224 997009407 43741184
10 P Windows Recovery Env 997011456 1000214527 3203072

P=Primary D=Deleted
>[Quick Search] [Backup]
Try to locate partition
```

The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for partitions

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TestDisk displays the first results in real time.



(click on thumb to display the image).

During the **Quick Search**, TestDisk has found two partitions including the missing logical partition labeled **Partition 3**.

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https://www.cgsecurity.org

Disk /dev/nvme0n1 - 512 GB / 476 GiB - CHS 488386 64 32
Partition      Start      End      Size in sectors
P EFI System    2048      616447    614400 [EFI System Partition] [ESP]
D MS Data       878592    778887167 778008576 [OS]
D MS Data       878592    783083519 782204928 [OS]
D MS Data       878592    950857727 949979136 [OS]
D MS Data       878592    997011455 996132864 [OS]
D MS Data       774692865 778887168 4194304
D MS Data       778887168 783081471 4194304 [New Volume]
D EFI System    783083520 785084415 2000896 [EFI System Partition] [NO NAME]
D Linux filesystems. data 785084416 942192639 157108224
D MS Data       909527041 953268224 43741184
D Linux Swap    942192640 950857711 8665072
D MS Data       948447233 950857728 2410496
D MS Data       950857728 953268223 2410496 [WINRETOOLS]
D MS Data       953268224 997009407 43741184 [Image]
D MS Data       993808385 997011456 3203072
D MS Data       997011456 1000214527 3203072 [DELLSUPPORT]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
P=Primary D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
NTFS, blocksize=4096, 2147 MB / 2048 MiB
```

- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

All directories and data are correctly listed.

- Press Enter to proceed.

Save the partition table or search for more partitions?

```
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https://www.cgsecurity.org

Disk /dev/nvme0n1 - 512 GB / 476 GiB - CHS 488386 64 32

Partition      Start      End      Size in sectors
1 P EFI System    2048      616447    614400 [EFI System Partition] [ESP]

[ Quit ] [ Return ] >[Deeper Search] [ Write ]
Try to find more partitions
```

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- **When all partitions are available** and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu **Extd Part** gives you the opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.
- **Since a partition, the first one, is still missing**, highlight the menu **Deeper Search** (if not done automatically already) and press Enter to proceed.

A partition is still missing: Deeper Search

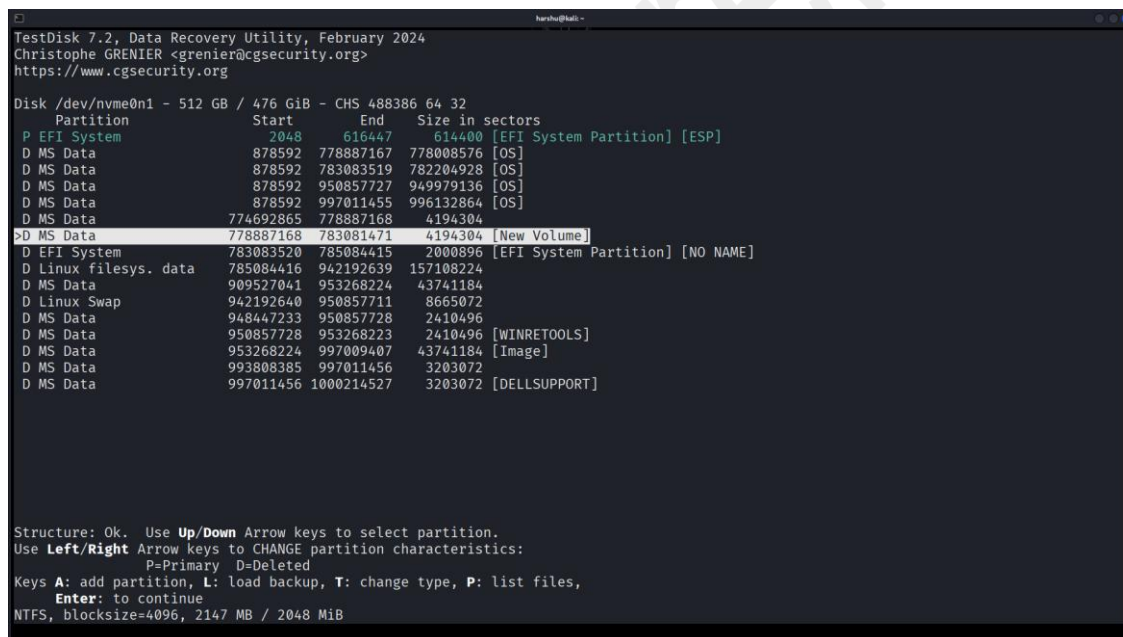
Deeper Search will also search for FAT32 backup boot sector, NTFS backup boot superblock, ext2/ext3 backup superblock to detect more partitions,

it will scan each cylinder  (click on thumb).

After the Deeper Search, the results are displayed as follows:

The first partition **"Partition 1"** was found by using backup boot sector. In the last line of your display, you can read the message **"NTFS found using backup sector!"** and the size of your partition. The "partition 2" is displayed twice with different size.

Partitions listed as D(eleted) will not be recovered if you let them listed as deleted. Both partitions are listed with status **D** for deleted, because they overlap each other. You need to identify which partition to recover.



```
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https://www.cgsecurity.org

Disk /dev/nvme0n1 - 512 GB / 476 GiB - CHS 488386 64 32
Partition      Start      End      Size in sectors
P EFI System    2048      616447    614400 [EFI System Partition] [ESP]
D MS Data       878592    778887167 778008576 [OS]
D MS Data       878592    783083519 782204928 [OS]
D MS Data       878592    950857727 949979136 [OS]
D MS Data       878592    997011455 996132864 [OS]
D MS Data       774692865 778887168 4194304
D MS Data       778887168 783081471 4194304 [New Volume]
D EFI System    783083520 785084415 2000896 [EFI System Partition] [NO NAME]
D Linux filesystems. data 785084416 942192639 157108224
D MS Data       909527041 953268224 43741184
D Linux Swap    942192640 950857711 8665072
D MS Data       948447233 950857728 2410496
D MS Data       950857728 953268223 2410496 [WINRETOOLS]
D MS Data       953268224 997009407 43741184 [Image]
D MS Data       993808385 997011456 3203072
D MS Data       997011456 1000214527 3203072 [DELLSUPPORT]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
P=Primary D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
NTFS, blocksize=4096, 2147 MB / 2048 MiB
```

- Highlight the first partition Partition 2 and press **p** to list its data.

The file system of the upper logical partition (label Partition 2) is  (click on thumb).

- Press q for Quit to go back to the previous display.
- Let this partition Partition 2 with a damaged file system marked as **D**(deleted).
- Highlight the second partition Partition 2 below
- Press p to list its files.

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```
TestDisk 7.2, Data Recovery Utility, February 2024
Please select a destination where /4Dell Latitude CPi.E01 will be copied.
Keys: Arow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /home/harshu
drwx----- 1000 1000 4096 24-Aug-2026 07:46 .
drwxr-xr-x 0 0 4096 17-Jun-2026 16:11 ..
drwxrwxr-x 1000 1000 4096 17-Jun-2026 23:30 4-ZERO-3
>drwxr-xr-x 1000 1000 4096 1-Aug-2026 05:17 Desktop
drwxr-xr-x 1000 1000 4096 17-Jun-2026 16:12 Documents
drwxr-xr-x 1000 1000 4096 24-Aug-2026 07:11 Downloads
drwxr-xr-x 1000 1000 4096 17-Jun-2026 16:12 Music
drwxr-xr-x 1000 1000 4096 24-Aug-2026 07:47 Pictures
drwxr-xr-x 1000 1000 4096 17-Jun-2026 16:20 Projects
drwxr-xr-x 1000 1000 4096 17-Jun-2026 16:12 Public
drwxr-xr-x 1000 1000 4096 17-Jun-2026 16:12 Templates
drwxr-xr-x 1000 1000 4096 17-Jun-2026 16:12 Videos
drwxrwxr-x 1000 1000 4096 18-Jun-2026 02:22 back-me-up
drwxr-xr-x 1000 1000 4096 29-May-2026 10:26 go
drwxrwxr-x 1000 1000 4096 18-Jun-2026 04:23 loxs
drwxrwxr-x 1000 1000 4096 24-Jun-2026 11:40 nasa
drwxrwxr-x 1000 1000 4096 18-Jun-2026 02:29 reports
drwxrwxr-x 1000 1000 4096 24-Jun-2026 09:02 subzy
drwxrwxr-x 1000 1000 4096 17-Jun-2026 23:29 testssl.sh
drwxrwxr-x 1000 1000 4096 16-Jul-2026 11:02 volatility3
-rw-r--r-- 0 0 671094597 7-Aug-2026 05:49 4Dell Latitude CPi.E01
-rw-rw-r-- 1000 1000 66861266 17-Jun-2026 23:14 go1.26.4.linux-amd64.tar.gz
-rw-rw-r-- 1000 1000 2354521 24-Jun-2026 09:03 nasa_subdomains
-rw-rw-r-- 1000 1000 0 24-Jun-2026 09:20 subdomain_alive.txt
-rw-r--r-- 0 0 13125 24-Aug-2026 07:51 testdisk.log
```

It works, your files are listed, you have found the correct partition!

- Use the left/right arrow to navigate into your folders and watch your files for more verification

Note: FAT directory listing is limited to 10 clusters - some files may not appear but it doesn't affect recovery.

- Press q for Quit to go back to the previous display.
- The available status are Primary, * bootable, Logical and Deleted.

Using the left/right arrow keys, change the status of the selected partition from D(eleted) to **L(ogical)**. This way you will be able to recover this partition.

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Hint: read [How to recognize primary and logical partitions?](#)

Note: If a partition is listed *(bootable) but if you don't boot from this partition, you can change it to **Primary** partition.

- Press Enter to proceed.

Partition table recovery

It's now possible to write the new partition structure.

Note: The extended partition is automatically set. TestDisk recognizes this using the different partition structure.

```
TestDisk 7.2, Data Recovery Utility, February 2024
Please select a destination where /4Dell Latitude CPi.E01 will be copied.
Keys: Arow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /home/harshu/Desktop
>drwxr-xr-x 1000 1000 4096 1-Aug-2026 05:17 .
drwx----- 1000 1000 4096 24-Aug-2026 07:46 ..
-rw-rw-r-- 1000 1000 70 18-Jun-2026 10:21 storage
-rw-rw-r-- 1000 1000 445 19-Jun-2026 15:11 subdomains.txt
```

- If **all partitions are listed** and only in this case, confirm at **Write** with Enter, y and OK.

Now, the partitions are registered in the partition table.

NTFS Boot sector recovery

The boot sector of the first partition named **Partition 1** is still damaged. It's time to fix it. The status of the NTFS boot sector is bad and the backup boot sector is valid. Boot sectors are not identical.

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```
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Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
MS Data 778887168 783081471 4194304 [New Volume]
Directory /4Dell Latitude CPi.E01
Copy done! 1 ok, 0 failed
dr-xr-xr-x 0 0 0 24-Aug-2026 04:49 .
dr-xr-xr-x 0 0 0 24-Aug-2026 04:49 ..
dr-xr-xr-x 0 0 0 7-Aug-2026 06:09 $RECYCLE.BIN
dr-xr-xr-x 0 0 0 24-Aug-2026 04:55 EXP-5
dr-xr-xr-x 0 0 0 7-Aug-2026 07:09 System Volume Information
>-r--r--r-- 0 0 671094597 7-Aug-2026 05:49 4Dell Latitude CPi.E01
-r--r--r-- 0 0 243 7-Aug-2026 05:49 4Dell Latitude CPi.E01:Zone.Identifier

Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'c' to copy the selected files, 'c' to copy the current file
```

- To copy the backup of the boot sector over the boot sector, select **Backup BS**, validate with Enter, use y to confirm and next OK.

More information about repairing your boot sector under [TestDisk Menu Items](#). The following message is displayed:

The boot sector and its backup are now both OK and identical: the NTFS boot sector has been successfully recovered.

- Press Enter to quit.